Mohammed Elghanam

Global Engineering Project Portfolio

R&D CTO | Embedded Systems, IoT, Smart Utilities, LPWAN, FPGA & Safety-Critical Systems

A curated selection of engineering programs, embedded platforms, industrial systems, and product development projects delivered across utilities, industrial automation, communications, medical device R&D, and mission-critical applications.

- Embedded Systems
- Smart Utilities
- Industrial Automation
- Mission-Critical Platforms

International Recognition | IIFME Kuwait

★ 5 Granted Patents

Engineering Consultant | Technical Expert | Open to consulting, partnerships, and technology collaboration opportunities

Engineering Portfolio Overview

Executive Profile & Capability Map

Mohammed Elghanam is an R&D CTO, Engineering Consultant, and Technical Expert with 25+ years of experience in embedded systems, industrial electronics, and product development. His work spans smart utilities, industrial automation, communications, medical device R&D, and mission-critical engineering.

25+
Years Experience

5
Granted Patents

Target Markets:
Egypt | GCC | Remote Global

CAPABILITY MAP

- Embedded Systems Architecture
- IoT & LPWAN Product Development
- Smart Metering & AMI
- FPGA-Based Design
- Hardware/Firmware Integration
- RTU & Industrial Automation
- Power Protection & Control
- Safety-Critical Systems
- Mission-Critical Communications
- Technical Strategy & R&D Leadership

ROLE SCOPE

R&D Leadership
Leading research, innovation, and advanced engineering initiatives.

Solution Architecture
Designing scalable, secure, and reliable engineering solutions.

Engineering Consulting
Providing expert consulting and problem-solving across complex systems.

Technical Advisory
Advising on technology roadmaps, standards, and strategic decisions.

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Smart Utilities & Smart Metering Solutions

ENGINEERING INTELLIGENCE. CONNECTED COMMUNITIES. SUSTAINABLE IMPACT.

CONNECT • MEASURE • MANAGE • BILL • EMPOWER

ENGINEERING CHALLENGE

Modernize electricity, water, and gas metering while supporting legacy infrastructure and scalable deployment.

ENGINEERED SOLUTION

- Developed prepaid and smart metering solutions for electricity, water, and gas.
- Upgraded legacy meters to support advanced metering capabilities.
- Integrated LPWAN, RS485, cellular, Wi-Fi, and PLC communications.
- Built scalable data management and billing capabilities.

MY ROLE

- Led hardware design for utility meter families.
- Supervised integration between hardware and software teams.
- Guided system integration and implementation.
- Directed communication protocol strategy.

APPLICATION CONTEXT

Utility modernization and rural community deployment, including the Egyptian Decent Life Initiative.

VALUE DELIVERED

- Improved revenue collection
- Enhanced customer service
- Increased operational efficiency
- Supported more efficient resource usage

Patented LPWAN IoT Platform

A Flagship R&D Project

ENGINEERING CHALLENGE

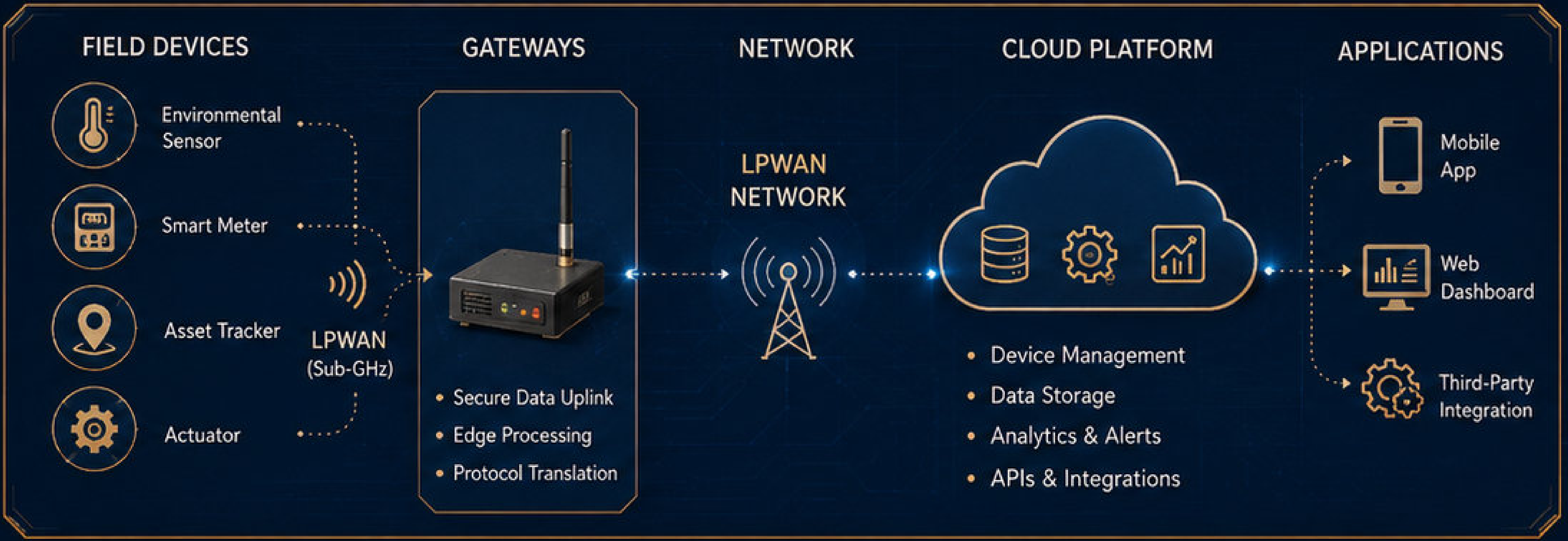
Create a long-range, low-power, cost-effective IoT platform suitable for diverse field deployments.

ENGINEERED SOLUTION

Developed an end-to-end LPWAN IoT platform covering field devices, gateways, embedded software, cloud connectivity, and mobile applications.

APPLICATIONS

- Smart Lighting
- Waste Management
- Agriculture
- Smart Homes / Buildings



MY ROLE

- Spearheaded the R&D effort from concept to deployment
- Led system architecture and technology strategy
- Aligned hardware and software development
- Secured intellectual property

INNOVATION HIGHLIGHTS

- Long-range connectivity (up to several kilometers)
- Low-power operation for extended battery life
- Scalable architecture for massive device deployments
- Commercial applicability with cost-effective design

OUTCOME

- Technological innovation with patented IP
- Market-ready platform addressing real-world needs
- Positive economic and social value through digital transformation

RTU & Industrial Automation Systems

RELIABLE. INTEROPERABLE. SECURE. BUILT FOR INDUSTRY.

PROJECT FOCUS

Advanced universal standard RTU sets for industrial monitoring and control.

ENGINEERING CHALLENGE

Deliver reliable, interoperable, and secure RTU platforms for industrial environments.

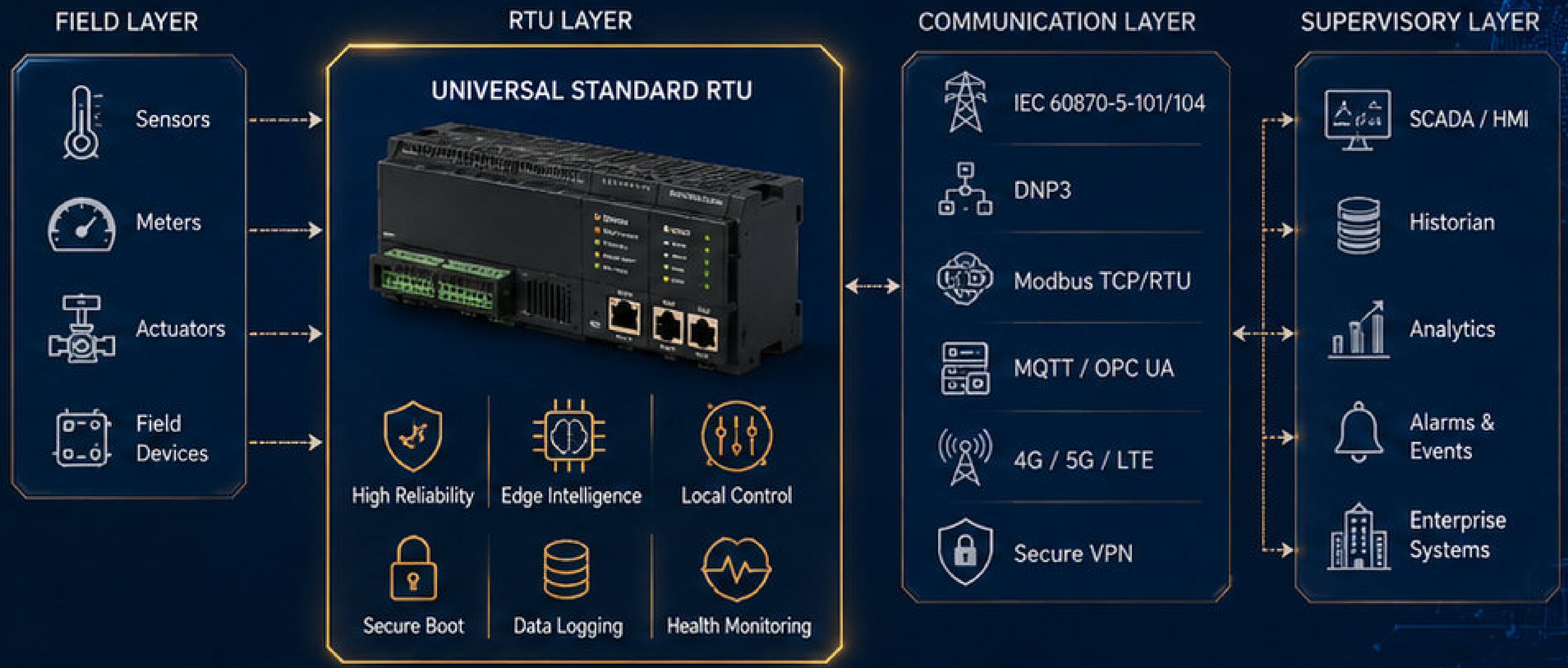
ENGINEERED SOLUTION

Supervised end-to-end architecture definition, hardware/firmware integration, validation, and industrial communication implementation; leveraged embedded energy management systems to build scalable, energy-efficient platforms.

MY ROLE

Led multidisciplinary engineering coordination, guided architecture decisions, ensured system validation, and oversaw secure interoperable operation.

END-TO-END RTU ARCHITECTURE



SECURITY BY DESIGN | Encryption in Transit | Role-Based Access Control | Secure Firmiare Updates | Audit & Logging

TECHNICAL THEMES

- INDUSTRIAL MONITORING
- CONTROL INTEGRATION
- COMMUNICATION PROTOCOLS
- CYBERSECURITY CONSIDERATIONS
- PLATFORM SCALABILITY

VALUE DELIVERED

Contributed to industrial automation efficiency and digital transformation readiness.

Power Grid
Protection & Control Systems

Modernizing Power Grid Protection with
Digital Overcurrent & Earth Fault Relay for 11kV Networks

CLIENT CONTEXT
Middle East Electricity Company

ENGINEERING CHALLENGE
Improve protection accuracy, reliability, and integration with an existing control environment.

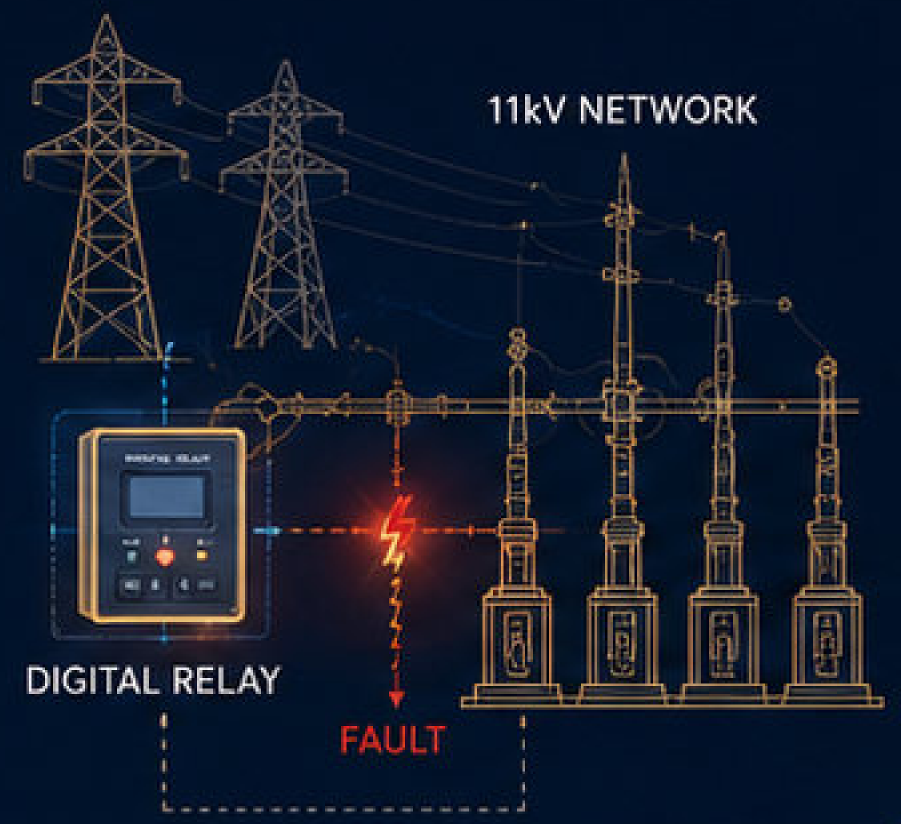
ENGINEERED SOLUTION
- Designed and developed a digital overcurrent and earth fault relay.
- Implemented fast fault detection and isolation.
- Integrated the solution with the Foxboro Distributed Control System (DCS).

MY ROLE
- Led hardware and software development.
- Directed testing for compliance with industry and IEC requirements.
- Managed installation and commissioning at substations.

RESULTING VALUE
- Improved power system reliability.
- Reduced service interruptions.
- Enhanced safety for people and equipment.

DIGITAL OVERCURRENT & EARTH FAULT RELAY

- FAST FAULT DETECTION
- ACCURATE PROTECTION
- SELECTIVE ISOLATION
- IEC 60255 COMPLIANT



INTEGRATION WITH FOXBORO DISTRIBUTED CONTROL SYSTEM (DCS)

11kV Networks | Digital Relay | IEC-Oriented Testing | Foxboro DCS Integration

Mission-Critical Communication Platforms

End-to-End Engineering Excellence for Secure, High-Performance Systems

01 Delta Multiplexing Communication Platform

Supervised full lifecycle development and integration, including component design, FPGA-based implementation, optimization for performance, power efficiency, and secure operation.

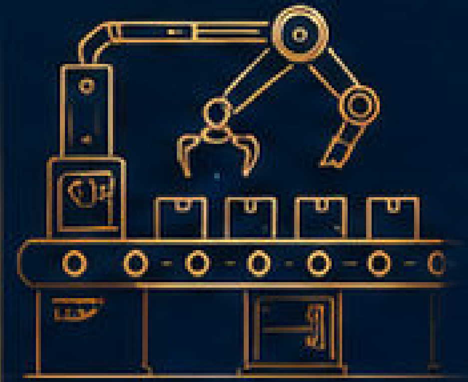
02 Encryption/Decryption Communication System

Oversaw architecture, MMI integration, FPGA-based implementation, and full testing and system integration.

03 SDR Prototype Hardware Review

Provided hardware design expertise for a software-defined radio prototype; reviewed the main board with ARM processor, Xilinx FPGA, and DDR3 RAM, reviewed the RF board, and applied signal integrity and EMC guidelines.

04 Delta Switching TDMA Production Support



Managed and supervised mass production activity in cooperation with an industrial partner.

My Contribution Across Programs

- Lifecycle supervision
- Architecture guidance
- FPGA optimization
- Hardware review
- Integration oversight
- Validation direction

Secure by Design | Reliable by Engineering | Delivered with Integrity

Trusted Solutions for Critical Communication Systems

Unified Azan System

Coordinated. Accurate. Unified.

PROJECT AIM

Harmonize the call to prayer across Cairo, Qaliubia, and Giza.

ENGINEERING CHALLENGE

Coordinate accurate timing, reliable signal distribution, and integration with existing mosque systems at scale.

ENGINEERED SOLUTION

Developed the technical infrastructure using RDS, GPS-based timing systems, high-powered amplifiers, and robust communication networks.

MY ROLE

Architected the system, directed the development team, and resolved complex technical issues related to synchronization, signal propagation, and integration.

WHY IT MATTERS

Delivered a coordinated, scalable system for reliable synchronized broadcasting.

GPS
Precise Timing Synchronization

BROADCAST
RDS Transmission Wide Coverage

TIMING
Accurate Azan Scheduling

NETWORK
Reliable Communication & Orchestration

Serving Communities | Strengthening Unity
Technology in service of faith and society.

SMART UTILITIES
CIVIC TECHNOLOGY SOLUTIONS

Medical Device R&D Systems

Human-Centered Engineering for Safer, Accessible Healthcare

Engineering Focus

Develop accessible medical support systems with safe control behavior, practical performance, and affordability.

MY ROLE

Spearheaded the projects from design and prototyping through control development and testing.

01 ARTIFICIAL VENTILATOR SYSTEM

- Precise control of tidal volume, respiratory rate, and I/E ratios.
- Design, prototyping, control algorithms, and safety-oriented testing.

02 OXYGEN CONCENTRATOR SYSTEM

- Concept development, control algorithms, prototyping, and testing focused on accessible oxygen therapy.
- Reliable performance with user-friendly operation and safety in focus.

VALIDATION & CERTIFICATION

Both systems received calibration and safety certification from the Cairo University Medical Calibration and Safety Center.

Official marketing and production licensing with the Egyptian Drug Authority is in progress.

Cairo University
Medical Calibration and Safety Center

EDA
EGYPTIAN DRUG AUTHORITY

IMPACT

Improved healthcare accessibility and affordability potential, supporting better outcomes for more patients.

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Railway Safety, Recognition & Contact

Engineering Safe Journeys. Earning Trust. Delivering Impact.

Railway Level Crossing Control System

Project Context

Factory 909, Ministry of Military Production.

My Role

Key subcontractor for design, development, installation, and commissioning; contributed to signal processing units and safety interlocks while ensuring adherence to strict safety standards.

Value Delivered

Enhanced railway safety, improved operational efficiency, and strengthened public safety.

Recognition & Trust Signals

5 Granted Patents

International Recognition at IIFME Kuwait for the invention *"Upgrading of Traditional Meters into Smart Meters"*

REPUBLICA PORTUGUESA

INVENTARIUM

APIICIS

N. 016 - 16

OFFICIAL CERTIFICATE

IN SPECIAL SCIENTIFIC MERIT
AND EXCELENT INTERNATIONAL COOPERATION

IT CERTIFIES THAT IS GRANTED TO

ENG. MOHAMMED ELGHANAM

FROM

EGYPT

FOR THE EXCELLENT INVENTION

"UPGRADING OF TRADITIONAL METERS INTO SMART METERS"

FOR ALL THE ABNEGATED SHARE, COOPERATION
AND SPARKLING OPEN MIND
FOR THEIR FOR NEW IDEAS
PROMOTING INNOVATION & CREATIVITY
IN THE INVENTIVE SPIRIT OF PROGRESS AND
INTERNATIONAL COOPERATION BETWEEN COUNTRIES

IIFME 16 – KUWAIT * The International Invention Fair of the Moddle East
January 29 to February 1, 2016

Manuel Maria Potes de Cordovil
Vice - President

AEI

Fernando Maldonado Lopes
President

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